Gravitational field strength

antocciphysics | TEACHER KEY | CPA

Learning target: Use a gravitational field to predict the force on a test mass.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 3; 3 course days later.*
Mass 3 kg; weight 15 N toward the source. Weight depends on field.

02. Explain a worked example (worked)
Field 3 N/kg remains the same; forces differ with test mass and point along the field.

03. Look and explain (model)
Arrows toward source; acceleration follows local inward field.

04. Build the solution (practice)
4, 8, 12 N; field remains 4 N/kg.

05. Use the idea (reason)
Force grows with mass but dividing by same mass gives a=g; ideal acceleration independent of test mass.

06. Choose a method (mixed) *Review: Unit 1, Day 16; 12 course days later.*
Net 32 N right; acceleration 8 m/s² right. Leftward motion initially slows.

07. Exit check (exit)
g 5 N/kg; 20 N.

08. STRETCH: Apply it another way (stretch)
Example 1 and 3 kg give 5 and 15 N. Field does not triple when test mass triples.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 3; 3 course days later.*
Mass 3 kg; new weight 15 N.

02. Explain a worked example (worked)
Field 3 N/kg remains the same; forces differ with test mass and point along the field.

03. Represent and interpret (model)
Arrows toward source; acceleration follows local inward field.

04. Solve and justify (practice)
g 3 N/kg; $r=\sqrt{(12/3)}=2$ m.

05. Analyze the reasoning (reason)
Force grows with mass but dividing by same mass gives a=g; ideal acceleration independent of test mass.

06. Choose a method (mixed) *Review: Unit 1, Day 16; 12 course days later.*
Downslope weight 19.6 N; net 8 N down; a=2 m/s² down. Normal force≈33.95 N.

07. Exit check (exit)
2 N/kg; 1 N toward source.

08. STRETCH: Apply it another way (stretch)
At 2 r; force 0.5 N toward the source.